

Evaluation Pipeline

Note

- AP — Area under Precision Recall Curve — per class per cell.
- mAP — mean of APs of all classes per cell.
- mAP@[.5:.95] — mean of mAPs across various IoU thresholds.

(Each IoU threshold gives one precision recall curve.)

Layerwise Metric Breakdown

- **Layer A — Per (Class, Cell)** (*Evaluated individually per class*)
 - AP@[.5:.95]
 - N (Ground-truth instance count)
- **Layer B — Per Cell (Task Summary)** (*Macro-averaged over 4 frozen core classes: car, sign, light, person*)
 - mAP@[.5:.95] (Primary task scalar)
- **Layer C — Per Client (Temporal)**
 - Final mAP
 - Forgetting
 - Backward Transfer (BWT)
- **Layer D — System Level**
 - Average mAP
 - Average Forgetting
 - Average BWT
 - Client Spread (Min, Max, Std)
- **Layer E — Cross-Client Transfer**
 - 🏆 **Global Cross-Client mAP (HEADLINE METRIC)**
 - 5×5 Transfer Matrix (mAP@[.5:.95])
 - Own-vs-Other Gap
- **Layer F — Spatial Interference Matrix**
 - Spatial Forgetting ($\text{mAP}_{\text{own}}(\text{before}) - \text{mAP}_{\text{own}}(\text{after})$)

Categorical Split

- **Per-Client Metrics**

- `AP@[.5:.95]`
- `N` (Ground-truth instance counts)
- `mAP@[.5:.95]` (Primary task scalar)

- **Continual Metrics**

- `Final mAP`
- `Forgetting` ($\text{mean}_j(\max(\text{history}_j) - \text{final}_j)$)
- `Backward Transfer (BWT)` ($\text{mean}_j(\text{final}_j - \text{first}_j)$)

- **Federated Metrics**

- `Spatial Forgetting` (5×5 pairwise gossip interference matrix)
- `$5 \times 5$  Transfer Matrix` (`mAP@[.5:.95]`)
- `Own-vs-Other Gap` ($\text{mean}(\text{diagonal}) - \text{mean}(\text{off-diagonal})$)

- **Global Metrics**

- 🏆 **Global Cross-Client mAP (HEADLINE METRIC** — Off-diagonal mean of Layer E matrix)
- `Average mAP` (System-wide mean)
- `Average Forgetting` (System-wide mean)
- `Average BWT` (System-wide mean)
- `Client Spread` (Min, Max, Std across clients)

Training Pipeline

Major Changes

1. **Momentum reset after gossip**
 - Momentum is reset after each gossip operation because the model parameters may change.
2. **Separate learning rates**
 - **New classifier head:** LR = 0.01
 - **Pretrained components:** LR = 0.001, or potentially frozen completely.
3. **Post-training gossip participation**
 - A client that has completed its local training can still **share its weights** through gossip.
 - Its own weights are **not modified** by the gossip operation.
 - This allows other clients to benefit from its learned knowledge.
4. **Spatial interference evaluation**
 - **Spatial probe size:** 200 images
 - **Spatial probe interval:** Every 100 gossip operations
 - The 200-image probe is used to evaluate model performance **before and after gossip**, allowing spatial interference to be measured.
5. Increased epoch to 3, so that the new classifier head can learn fast.

2.1 Gossip Parameters

Parameter	Configuration
Replay ratio	1 — current-task samples : replay samples = 1:1 (50–50)
Communication interval	Every global round; one gossip operation per global round
Buffer capacity	2000 images
Gate loss	Same as the original model loss
Gate sensitivity	2
Gate comparison batch size	64
Gate chunk size	64
Gossip λ	0.5

2.1.1 Buffer Overflow Policy

When the replay buffer is full and images from the current task need to be added:

- Identify images based on the **rarest class** present in each image.
- Images containing rarer classes receive **higher priority**.

- During removal, prioritize removing images with more common classes.
- Images containing rare classes are preferentially **retained** in the buffer.

2.2 Federated Continual Learning Training Pipeline

Component	Configuration
Base model	COCO-pretrained Faster R-CNN
Frozen components	ResNet backbone, FPN, RPN
Trainable component	New classifier head
Number of classes	11 classes, including background
Epochs	3
Batch size	32
AMP	Enabled
Normalization	Built-in Faster R-CNN normalization
Optimizer	SGD
Momentum	0.9
Momentum reset	Reset after gossip, since model parameters change
Learning rate — new head	0.01
Learning rate — pretrained components	0.0001 / potentially frozen completely
Data augmentation	None
Loss function	Default Faster R-CNN loss
Total loss	Classification loss + box regression loss + objectness loss + proposal box regression loss
Learning-rate warm-up	None
BK	None

2.2.1 Gossip mechanism

- After completing local training, a client **continues to participate in gossip**.
- The client **shares its learned knowledge/weights** with other clients.
- However, the client **does not modify its own weights as a result of gossip**.
- Momentum is **reset after gossip** because the received model information changes the model state.